Exact replication and screening of tropical finiteness certificates for central configurations

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Abstract

We report a replication and screening study around Smale's 6th problem (finiteness of planar central configurations). Conclusions rest on exact rational or integer arithmetic, with three stated exceptions: the `gfan` prevariety runs use the machine-integer fast path (the overflow-checked rerun has not been made), the random linear section probes are probabilistic, and the modular Gröbner screens are used for triage only. The contributions are: (i) an exact calibration of the Albouy–Chenciner mutual-distance equations in open tooling, including a machine rediscovery of their dimension-blindness (the regular tetrahedron coexists with the square in the equal-mass rhombus stratum of the 4-body distance system until the Cayley–Menger equation is adjoined; under the normalization that fixes the force constant to the total mass, the square, with side a satisfying $a^3 = (4+\sqrt{2})/8$, the larger of the two positive roots of $32a^6 - 32a^3 + 7 = 0$, is then the unique positive point of the stratum); (ii) a machine measurement of the *enrichment law*: adjoining the asymmetric (Roberts) equations and the energy–inertia relation renders the $n = 3$ distance ideal zero-dimensional directly; (iii) an independent reproduction of the Jensen–Leykin tropical prevariety certificate of generic-mass finiteness for $n = 5$ (arXiv:2301.02305v2): both published f-vectors match digit for digit and the pointedness of the recession cone is verified by an independent exact parser, on the fast path; the same binary in arbitrary precision reproduces the three $n = 4$ f-vectors it shares with the screening; (iv) a screening over mass-valuation families and equation-system variants at $n = 4, 5$ with complete tables, which finds two new working valuation families beyond the two published, replicates generic-mass finiteness for $n = 4$ purely polyhedrally in seconds, and shows that removing the ten dependent symmetric Albouy–Chenciner equations destroys the certificate at every tested valuation; every comet is decided exactly by a rational-arithmetic feasibility certificate, and adding valid equations is measured separately (Dziobek's equations are tropically active but not decisive at the hard case; the energy–inertia relation is inert); (v) a status report on the two-pair reflection-symmetric stratum of the six-body problem: the cases $k = 0, 1, 2, 4$ of a fibre-dimension chain are proved with written proofs, the case $k = 3$ is reduced to four bilinear conditions, an interval atlas certifies every region except one exact degenerate point (the cross point) and

one chart region still open, the degenerate set of the cross family is finite, and no finiteness statement is claimed for the stratum; and (vi) an audit of the Lean formalization of the five-body finiteness claims: its main theorem states exactly the classical claim, and a scan of nine kernel-bypass patterns over 2,751 stored and 85,756 generated modules finds none of them, which does not replace the kernel check. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin and Jensen–Leykin; the measurements in (i), (ii) and (iv) are, to our knowledge, new at this level of explicitness, a statement not confirmed by a systematic literature search.

1 Introduction

A *central configuration* (CC) of the Newtonian n-body problem with masses $m_1, \dots, m_n > 0$ is a configuration $x = (x_1, \dots, x_n)$, $x_i \in \mathbb{R}^d$, satisfying

$$\lambda\,(x_i - c) = \sum_{j \neq i} \frac{m_j (x_j - x_i)}{r_{ij}^3}, \qquad i = 1, \dots, n, \tag{1}$$

for some λ, where c is the center of mass and $r_{ij} = |x_i - x_j|$. Smale's 6th problem for the 21st century [4] asks, following Chazy [5] and Wintner [6], whether for every n and every choice of positive masses the number of *planar* central configurations, up to symmetry, is finite. The problem text was read in the author's second version of 7 August 1998; the journal printing was not read [V].

Evidence marks. Statements carry one of four marks. [MV] marks a statement machine-verified in exact arithmetic, or by a recorded check that can fail; [D] a statement derived in the text or in a written derivation held in the experiment records; [V] a statement read on the primary source; [U] a statement not confirmed on a primary source. A statement is called *proved* only when the paper gives its proof or points to a written proof in the records by file and line range; otherwise it is stated as the records state it (reduced, claimed, open, dropped). Records are cited relative to `problems/dynamical-systems/central-configurations/`, abbreviated `CC/`; `EXP-NNN` names an experiment folder in `CC/experiments/`.

The ladder. For $n = 4$ the number is finite for all positive masses (Hampton–Moeckel [7]: computer-assisted BKK certificate, count in $[32, 8472]$ [V]). For $n = 5$ it is finite except possibly on an explicit codimension-two subvariety of mass space (Albouy–Kaloshin [8] [V], from the abstract and pp. 535–540). Finiteness for $n = 5$ and *all* positive masses was claimed in October 2026 by three groups, one with a Lean formalization (Section 11) [V]; none of the claims is refereed. The spatial 5-body Dziobek case is handled generically by Moeckel [9] and explicitly by Hampton–Jensen [10] [V]. The case $n = 6$ is open, reduced by Chang–Chen [11] to twenty-four unsolved zw-diagram cases (their programme preprint states the count verbatim, read on pages 1 to 6 [V]); for *generic* masses the purely polyhedral certificate of Jensen–Leykin [12] [V], complete through $n = 5$, is the live frontier, and a six-body symmetric stratum is treated in Section 9. Rigorous equal-mass censuses for $n \leq 7$ are due to Moczurad–Zgliczyński [13] [V]; with one negative mass a continuum exists (Roberts [15] [V], masses $(1, 1, 1, 1, -1/4)$), so positivity is essential.

Scope. This is a replication and screening study: it places the certificates of this ladder inside one open, reproducible toolchain with stored artifacts, and then runs the experiments that the

Tool	Release and build	Recorded SHA-256
`gfan`	0.7, author tarball, two-line `#include <cstdint>` patch for gcc 13, GMP, cddlib; WSL2 Ubuntu 24.04	tarball and binary: not in the committed artifacts (UNVERIFIED)
`gfan`	0.8beta, author tarball, same patch; binary installed as `gfan08`	tarball `fa7884e5f317c50f`, binary `0f177e6a4f7829fc` (prefixes)
`msolve`	0.10.1, built in WSL	binary `be6ea2289f0fab9847bf` (prefix)
Singular	4.3.2, Ubuntu 24.04 package	binary `90ab699b7a28486944a1` (prefix)
`sympy`	1.14, Python 3.13 (exact layer)	not applicable

Table 1: Tools, builds and recorded hashes. The prefixes are the first hexadecimal digits of the SHA-256 values in `CC/experiments/EXP-005-n6-prevariety-pow3/run08.sh`, `EXP-006-msolve-census/verdict.md` and `EXP-012-partial-gb-engine-spike/hypothesis.md`. The experiment record for the $n = 5$ reproduction states that the `gfan` 0.7 tarball and binary hashes were recorded in an artifacts file; that file is absent from the repository tree, so the values are not reproduced here.

Cell (system S1, $n = 5$)	gfan input, SHA-256 prefix	gfan output, SHA-256 prefix
powers of 3	`632a79188e206f50`	`4ec34432160d2241`
squares	`4f69cdd0798298bd`	`c8500283d31e3891`
powers of 2	`301ba5356dc79f5a`	`37c3c5dd73f958d4`
primes	`057c9c0cdf2c4448`	`973139ba1972eaef`
arithmetic	`23b202924e3e9f9a`	`a0be2344ba7a485c`
repeated	`56abb113b0b70798`	`602d223ad321a5e7`

Table 2: SHA-256 prefixes (16 hexadecimal digits) of the `gfan` inputs and outputs of the $n = 5$ cells of system S1, read from the stored hash file.

frontier papers themselves call for. Since the five-body claims of 2026, the $n = 5$ computations are no longer a route to a new theorem; they remain a replication, a screening of the method, and the calibration for the open case $n = 6$. Negative and capped results are reported with the positive ones.

2 Instruments and provenance

2.1 Tool versions and recorded hashes

The hashes of the 16 `gfan` inputs and 16 outputs of the screening grid are stored in `CC/experiments/EXP-004-valuation-equation-screening/artifacts/hashes.txt` (Table 2 reproduces the six cells of system S1 at $n = 5$).

2.2 The prevariety pipeline

Algorithm 1 is the exact pipeline of the runs in Sections 5 and 6. The system generator is `gfan`'s own `_nbody` command; the flag `--alsosymmetric` adds the symmetric Albouy–Chenciner equations to the asymmetric ones and `--cayleymenger2` adds the planar Cayley–Menger poly-

Algorithm 1 Tropical prevariety of the n-body system for a valuation vector $v = (v_1, \dots, v_n)$.

Require: n, valuations $v_i \in \mathbb{Z}$, system S1 or S2, a wall-time cap

1: `gfan _nbody -N`n `--masses --alsosymmetric --cayleymenger2 > system.txt` ▷ S2: drop `--alsosymmetric`
2: Replace the ring header `Q[m1,...,m`n`,` by `Q(t)[` and every word `m`i by t^{v_i} ▷ masses enter the field $\mathbb{Q}(t)$ with valuation v_i
3: `gfan _tropicalprevariety --usevaluation -j30 --mint --minx --bits 64`, reading `input.txt` and writing `output.out` ▷ caps: 2400 s at $n = 5$, 600 s at $n = 4$
4: Read `F_VECTOR`, `RAYS` and `MAXIMAL_CONES` from the output
5: Count the unbounded directions (rays whose first coordinate is 0) that have a positive coordinate; zero means the whole recession cone is pointed (exact `Fraction` parser)

Ensure: f-vector, global pointedness flag, wall time

Algorithm 2 Comet extraction from a `gfan` prevariety output.

Require: `RAYS` and `MAXIMAL_CONES` of the output

1: Keep the maximal cones that contain at least one bounded ray (first coordinate not 0); the cones made only of unbounded rays have an empty slice
2: Union-find: for each kept cone, unite all of its *bounded* rays
3: Each class of bounded rays is a comet; attach to it the unbounded rays of every cone that contains one of its rays
4: The recession generators of the comet are the unbounded rays attached to it

Ensure: the list of comets with their generator sets

nomials. For $n = 5$ system S1 has 35 polynomials in $\mathbb{Q}[m_1, \dots, m_5, r_{12}, \dots, r_{45}]$ (20 asymmetric, 10 symmetric, 5 planar Cayley–Menger) [MV]. System S2 omits `--alsosymmetric`.

The runs of Section 6 that adjoin equations (the enrichment grid) use `-j8` and `--bits 0` (arbitrary precision) on the system of Algorithm 1 with the extra polynomials appended. In that mode the `gfan` 0.7 parser rejects an input that mixes t^0 with positive powers of t; because every polynomial of the system is homogeneous in the masses (the Albouy–Chenciner equations are linear in them, the energy–inertia relation is quadratic, the Cayley–Menger and Dziobek polynomials carry none), shifting all valuations by a constant multiplies each polynomial by a unit power of t and leaves the prevariety unchanged. The shifted arithmetic baseline reproduces the unshifted f-vector exactly [MV].

2.3 Comet extraction

A *comet* is a connected component of the slice $t = 1$ of the prevariety, with the recession cone generated by its unbounded rays. In the `gfan` output a ray with first coordinate not zero is a point of the slice and a ray with first coordinate 0 is an unbounded direction.

The connectivity rule is that cones are joined *only* through bounded rays. Sharing an unbounded ray does not join the slices at $t = 1$. A first version of the parser joined through unbounded rays as well; on the squares valuation at $n = 5$ it returned 245 fused components, and the corrected rule returns 257, the number of components reported in [12] [V] (`CC/experiments/EXP-004-valuation-equation-screening/verdict.md`, section on the second-level validation).

Algorithm 3 Exact pointedness of a comet (phase-I simplex over `Fraction`, Bland's rule).

Require: generators $g_1, \dots, g_k \in \mathbb{Q}^d$ ($d+1$, the number of rows, is 11 or 12 in our prevarieties)
1: Build the tableau of (2): $d+1$ rows, k structural columns, $d+1$ artificial columns; flip rows so that the right-hand side is ≥ 0
2: **repeat** choose the entering column as the smallest index with negative reduced cost (Bland); choose the leaving row by the minimum ratio, ties broken by the smallest basic index; pivot in rational arithmetic
3: **until** no negative reduced cost remains
4: **if** the phase-I optimum is 0 **then**
5: return UNPOINTED with λ; assert $\sum_i \lambda_i = 1$ and $G\lambda = 0$ exactly
6: **else**
7: take $c = -y$ from the phase-I dual; if $c \cdot g_i < 0$ for all i return POINTED with c; otherwise try $c = -\sum_i g_i$; if neither verifies, report POINTED without a certificate
8: **end if**
9: Independently of the decider, re-verify every certificate by exact substitution
Ensure: pointed or unpointed, with a verified certificate

2.4 The exact pointedness decider

A cone $C = \mathrm{cone}(g_1, \dots, g_k)$ in $\mathbb{R}^d$ is pointed exactly when the only nonnegative solution of $\sum_i \lambda_i g_i = 0$ is $\lambda = 0$. This is the single feasibility question

$$\exists\, \lambda \geq 0 : \quad G\lambda = 0, \quad \textstyle\sum_i \lambda_i = 1, \tag{2}$$

where G has the generators as columns. A feasible point is a certificate that the cone is not pointed; infeasibility yields, by Farkas' lemma and the phase-I dual, a vector c with $c \cdot g_i < 0$ for every generator, a certificate that it is pointed.

Bland's rule guarantees termination without cycling, and exact arithmetic makes each answer a proof rather than an estimate [D]. In the recorded run every one of the 16 cells was decided with zero certificate-verification failures; the unpointedness certificates are stored per cell (nine files in `CC/experiments/EXP-007-exact-pointedness-decision/artifacts/`) [MV].

2.5 Eliminant censuses and box containment

For an exact positive census at fixed rational masses we use two independent engines. The first (`sympy` over $\mathbb{Q}$) computes one graded reverse lexicographic Gröbner basis, forms univariate eliminants as characteristic polynomials of Stickelberger multiplication matrices, removes repeated factors, isolates the real roots, pre-filters numerically with certified bounds and accepts a point only after exact residual substitution of every defining polynomial; when the bare system is not zero-dimensional, a Rabinowitsch variable on the product of the distances saturates it. The second (`msolve` 0.10.1) computes a rational univariate representation and certified rational isolating boxes. The comparison is exact: containment of an exact algebraic point in a rational box is decided in exact arithmetic, each of our points must fall in a distinct box, and the number of boxes containing no point of ours (the unexplained boxes) must be zero [MV]. Algorithm 4 states the procedure.

Algorithm 4 Exact positive census at fixed rational masses, with box containment across two engines.

Require: the enriched distance system $F \subset \mathbb{Q}[r_{ij}]$ at fixed masses (with the Cayley–Menger equation when planarity is imposed)

1: If the system is not zero-dimensional, add a variable z and the relation $z\prod_{i<j} r_{ij} - 1$ (Rabinowitsch saturation)
2: **Engine A.** One graded reverse lexicographic Gröbner basis; the pure-power criterion on its leading terms decides zero-dimensionality
3: For each variable, take the characteristic polynomial of its multiplication matrix on the quotient ring (Stickelberger), remove repeated factors, isolate the real roots
4: Combine the roots into candidate points; pre-filter with certified numeric bounds; accept a point only when every defining polynomial vanishes exactly on it and all coordinates are positive
5: **Engine B.** `msolve -f input -o output` (rational univariate representation, certified rational isolating boxes); keep the boxes in the positive orthant, under a time cap of 1800 s at $n = 3$ and 3600 s at $n = 4$
6: For each accepted point of Engine A, decide exactly whether it lies in a box of Engine B
7: Accept the census when every point lies in a distinct box and no box is left without a point

Ensure: the positive census, or the report of the first disagreement

2.6 Section probes and the partial Gröbner bound

The probe asks whether the variety V of a system has dimension at most d. Integer coefficients are drawn uniformly from $[-10^6, 10^6]$ by a seeded generator, the draws are archived verbatim, and $d+1$ random affine-linear equations are adjoined; `msolve` then decides the zero-dimensional or empty system under a time cap. An empty answer (dimension -1) is evidence that no component of dimension at least $d+1$ exists; it is probabilistic, and a crude failure estimate of order 10^{-3} per draw follows from the Bézout ceiling of the stripped degrees [D]. The probe sees only the maximum dimension. The deterministic companion is the partial Gröbner bound of Dias–Pan [14]: the leading monomials of a Gröbner basis of any subideal belong to the leading-term ideal of the full ideal, so the union L of the leading monomials of tractable subideals bounds $\dim V$ above by the Krull dimension of $k[x]/(L)$ (a staircase computation). We compute the leading monomials with Singular over $\mathbb{Q}$ in degree reverse lexicographic order, which is sound as a bound; leading ideals printed by `msolve` are modular, so an unlucky prime could enlarge them and fake a lower bound, and they are used for screening only.

2.7 What is exact and what is not

Every claim of the form "this exact point satisfies this system", every comet decision, every interval certificate of Section 9 and every census at $n = 3$ rests on exact rational or integer arithmetic. The exceptions are: the `gfan` runs of Sections 5 and 6 (machine-integer fast path; Section 7); the section probes (probabilistic); the `msolve` modular leading ideals (triage only); and the completeness of `msolve`'s box list, which is trusted and cross-checked only against the classical counts at $n = 3$.

Algorithm 5 Dimension bounds for a variety V: random section probe and partial Gröbner bound.

Require: the defining polynomials of V, a target dimension d, a seed, time caps

1: **Probe.** With a seeded generator draw $d+1$ affine-linear equations with integer coefficients uniform in $[-10^6, 10^6]$; archive the draws verbatim
2: `msolve -P 2 -f input -o output` on the system plus the sections, under a cap (900 s at $n=4$, 300 s at $n=5$)
3: If the output is empty (dimension -1), record support for $\dim V \le d$ (probabilistic, failure estimate of order 10^{-3} per draw); a timeout supports nothing
4: **Partial Gröbner bound.** Choose tractable subideals $I_1, \dots, I_s \subset I(V)$; for each, compute $\mathrm{lead}(\mathrm{std}(I_j))$ in Singular over $\mathbb{Q}$ in degree reverse lexicographic order (cap 120 s)
5: Let L be the union of the leading monomials; $\dim V \le \dim k[x]/(L)$, computed by the staircase of L
6: Cross-check small cases against a control with known leading terms (a control caught a lexicographic leading-term error of the `sympy` engine)

Ensure: upper bounds on $\dim V$ with their strength: probabilistic for the probe, deterministic for the partial bound

3 Exact calibration of the Albouy–Chenciner system (EXP-001)

Working in the mutual distances r_{ij} with the scale normalization $S_{ij} = r_{ij}^{-3} - 1$ ($S_{ii} = 0$), the symmetric Albouy–Chenciner (AC) equations [16, 7] are, after clearing denominators,

$$f_{ij} = \sum_{k=1}^{n} m_k \left[S_{ik}\,(r_{jk}^2 - r_{ik}^2 - r_{ij}^2) + S_{jk}\,(r_{ik}^2 - r_{jk}^2 - r_{ij}^2) \right] = 0. \tag{3}$$

The normalization corresponds to the force constant λ equal to the total mass M (for four unit masses, $\lambda = 4$) [D]. Our exact implementation (sympy over $\mathbb{Q}$) reproduces, for four exact mass vectors, the classical $n=3$ landscape: the Lagrange point $r_{12} = r_{13} = r_{23} = 1$ identically in symbolic masses, and exactly one positive collinear solution per ordering (Euler–Moulton [17]; for equal masses the 2-middle chart value is $r_{12} = r_{23} = \sqrt[3]{90}/6$), with the degree-54 symbolic Euler eliminant persisted [MV]. For $n=4$ each cleared AC polynomial has exactly 14 monomials, matching the count stated in [7] [MV].

The instructive result of the calibration concerns the equal-mass rhombus stratum. In the stratum $r_{12} = r_{23} = r_{34} = r_{14} = a$, $r_{13} = r_{24} = b$ of $n=4$, the bare AC system has *two* positive solutions: the planar square, with $b^2 = 2a^2$ and $a^3 = (4+\sqrt{2})/8$ (minimal polynomial $32a^6 - 32a^3 + 7$), and the regular tetrahedron $a = b = 1$, whose bordered Cayley–Menger determinant equals $4 \neq 0$ [MV]. The distance equations are dimension-blind, exactly as stated in [7]; planarity is restored only by adjoining the Cayley–Menger equation, after which the square is the unique positive point of the stratum [MV].

The word *unique* needs its normalization. The sextic $32a^6 - 32a^3 + 7$ has two positive real roots, $a \approx 0.68628$ and $a \approx 0.87797$, and a square of four equal masses is central at every size, since scaling only rescales λ; the two roots give $\lambda \approx 8.3753$ and exactly 4. The root singled out is the larger one, the only one with the force constant of the normalization $\lambda = M = 4$ [MV] (`CC/context/2026-08-25-adversarial-audit.md`, section 2). We record the scale-free invariant

$J = UI^{1/2}/M^{5/2}$ at both anchors:

$$J_{\square} = \tfrac{1}{16} + \tfrac{\sqrt{2}}{8}, \qquad J_{\text{tet}} = \tfrac{3\sqrt{6}}{32} \tag{4}$$

(both exact) [MV].

Two toolchain findings of independent interest are covered by regression tests in the repository: `sympy`'s `solve_poly_system` silently returns incomplete solution lists on these systems (it missed the square, whose eliminant factor $4b^6-4b^3-7$ is even radical-solvable), and `is_positive` returns `None` on nested algebraic numbers, so naive filters silently drop valid solutions [MV]. Counting in our pipeline therefore uses univariate eliminants that are provably elements of the ideal (lex Gröbner elimination, or Stickelberger multiplication-matrix characteristic polynomials), complete real root isolation, and exact residual acceptance (Section 2).

4 The enrichment law (EXP-002)

The bare symmetric system is never zero-dimensional: the line $r_{13} = r_{23} = 0$ lies in its variety for all masses (machine-verified symbolically) [MV], and saturation is expensive. Adjoining the $n(n-1)$ asymmetric equations $g_{ij} = \sum_k m_k S_{ik}(r_{jk}^2 - r_{ik}^2 - r_{ij}^2)$ (observed by G. Roberts; see [10, 12]) and the energy–inertia relation $U = MI$ changes the picture completely: for all four sample mass vectors the enriched ideal is zero-dimensional *directly* (Gröbner basis of size 24 in under a second each), and the full positive censuses, where decided, are exactly classical: four points (one equilateral, three collinear), with no spurious solutions [MV]. The planar rhombus census on the enriched system with Cayley–Menger adjoined yields the square alone [MV]. This is the affine-level instance of a pattern that recurs at every level of the subject: Hampton–Moeckel could not run their method on the six AC equations alone and required Dziobek's equations [7]; Hampton–Jensen kept the asymmetric equations to refine their tropical cones [10]; and Jensen–Leykin explicitly call for "different versions of the Albouy–Chenciner equations" at $n = 6$ [12] [V]. Section 6 quantifies this pattern at the prevariety level.

5 Reproduction of the Jensen–Leykin $n = 5$ certificate (EXP-003)

Jensen–Leykin [12] prove generic-mass finiteness of planar CCs for $n \le 5$ by a purely polyhedral computation: specialize the masses into the Puiseux field with chosen valuations, compute the tropical prevariety of the system (20 asymmetric AC + 10 symmetric AC + 5 planar Cayley–Menger polynomials for $n = 5$) with `gfan` 0.7, and verify that every connected component has a pointed recession cone; Bieri–Groves and the density of tropicalization fibers then yield finiteness for almost all masses [V].

We rebuilt `gfan` 0.7 from the author tarball (one two-line `#include <cstdint>` patch for gcc 13) and ran the paper's own command pipeline (Algorithm 1). Results: with valuations $(1, 3, 9, 27, 81)$ the prevariety f-vector is $(1506, 4744, 8586, 8787, 4652, 993)$, and among its 85 unbounded ray directions none has a positive coordinate (verified by our independent exact parser): the globally pointed certificate, exactly as published [MV]. With valuations $(1, 4, 9, 16, 25)$ the f-vector is $(3586, 12012, 18531, 15625, 7072, 1357)$, again matching [12] digit for digit [MV]. Each run took under six minutes of wall time on 30 threads ($\approx$ 145 cpu-minutes; 5 m 52 s and 5 m 34 s). Both runs used the machine-integer fast path `--bits 64`, as in the timings of [12]; the overflow-checked arbitrary-precision rerun was not repeated (Section 7). To our knowledge

System	Valuations	comets	pointed	unpointed	gfan (s)	decider (s)
S1	$(1,3,9,27,81)$ powers of 3	281	281	0	225	0.9
S1	$(1,4,9,16,25)$ squares	257	257	0	362	2.2
S1	$(1,2,4,8,16)$ powers of 2	266	266	0	216	0.7
S1	$(2,3,5,7,11)$ primes	250	250	0	223	0.8
S1	$(0,1,2,3,4)$ arithmetic	181	180	1	379	1.4
S1	$(1,1,9,27,81)$ repeated	211	210	1	174	1.0
S2	$(1,3,9,27,81)$ powers of 3	233	232	1	87	1.5
S2	$(1,4,9,16,25)$ squares	206	205	1	115	2.2
S2	$(1,2,4,8,16)$ powers of 2	214	213	1	98	1.7
S2	$(2,3,5,7,11)$ primes	204	203	1	75	1.4
S2	$(0,1,2,3,4)$ arithmetic	139	138	1	107	2.2
S2	$(1,1,9,27,81)$ repeated	142	141	1	1253	15.8

Table 3: The twelve $n = 5$ cells (systems S1 and S2, six valuation families). Comets, pointed and unpointed counts and decider seconds from `EXP-007` (`summary.json`); f-vectors and `gfan` seconds from `EXP-004` (`screening-table.txt`). Every S1 cell with a repeated or arithmetic valuation has one unpointed comet; every S2 cell has exactly one. The squares row reproduces the 257 components reported in [12], a quantity independent of the f-vector (Table 4).

this is the first published independent replication of the [12] computation [U]; at the measured throughput, a single $n = 6$ attempt at the scale they report (≈ 100 cpu-days) costs about three wall-days on commodity hardware [D].

6 Valuation and equation-variant screening (EXP-004, EXP-007, EXP-008)

The certificate of [12] has two designed inputs: the mass valuations and the equation system. The paper publishes two working valuation choices for $n = 5$ and, for the inconclusive $n = 6$ attempt, asks explicitly for "more experimentation with different valuations or different versions of the Albouy–Chenciner equations" [V]. We screened both axes on a 16-cell grid: two systems (S1, the full set of asymmetric + symmetric AC equations + planar Cayley–Menger determinants; S2, the same with the $\binom{n}{2}$ dependent symmetric equations removed) crossed with six valuation families at $n = 5$, plus four cells at $n = 4$. Pointedness is decided per comet by an exact certificate (Algorithm 3). Tables 3 and 5 give the complete grid: f-vectors, comet counts, pointed and unpointed counts, and both times (the `gfan` wall time on the fast path and the decider time).

For the arithmetic and repeated valuations the heuristic screening reported "no certificate (one comet undecided)"; after the exact decision of Section 6.1 that comet is provably unpointed, and no comet of the grid is undecided.

Three findings.

(a) Two new working valuation families. Powers of two and primes both certify at $n = 5$ (266 of 266 and 250 of 250 comets pointed, Table 3); before this screening, two families were known to work. Since the certificate is a statement about one point of the valuation space, each working family is an independent verification of generic finiteness for $n = 5$, and the enlarged pool is what makes an $n = 6$ attempt a design choice rather than a lottery [D]. This matters operationally: powers of three reach t^{243} at $n = 6$ and exceed `gfan`'s machine-integer fast path

System	Valuations	f-vector
S1	$(1,3,9,27,81)$ powers of 3	1506 4744 8586 8787 4652 993
S1	$(1,4,9,16,25)$ squares	3586 12012 18531 15625 7072 1357
S1	$(1,2,4,8,16)$ powers of 2	2266 7285 11782 10702 5185 1044
S1	$(2,3,5,7,11)$ primes	2745 8795 13472 11540 5365 1059
S1	$(0,1,2,3,4)$ arithmetic	3985 13796 20950 17331 7663 1438
S1	$(1,1,9,27,81)$ repeated	1953 5151 5569 3039 898 109
S2	$(1,3,9,27,81)$ powers of 3	2839 9330 14440 12590 5984 1192
S2	$(1,4,9,16,25)$ squares	4425 15316 23071 18756 8250 1544
S2	$(1,2,4,8,16)$ powers of 2	3188 10801 16475 13798 6292 1216
S2	$(2,3,5,7,11)$ primes	3688 12406 18327 14780 6536 1240
S2	$(0,1,2,3,4)$ arithmetic	4439 16148 24682 20197 8838 1635
S2	$(1,1,9,27,81)$ repeated	1 7760 45625 116429 161123 126789 53611 9555 24

Table 4: The f-vectors of the twelve $n = 5$ cells of Table 3, from `EXP-004` (`screening-table.txt`). The first S1 cell and the squares cell match the two published f-vectors of [12] [V]. The S2 cell at repeated valuations has nine entries.

Valuations	comets	pointed	unpointed	gfan (s)	decider (s)	f-vector
$(1,3,9,27)$ powers of 3	10	10	0	4	0.0	44 80 57 14
$(1,2,4,8)$ powers of 2	10	10	0	2	0.0	63 119 82 19
$(0,1,2,3)$ arithmetic	9	9	0	2	0.0	83 164 108 23
$(0,0,0,0)$ equal	1	0	1	1	0.0	1 49 66 18

Table 5: The four $n = 4$ cells of system S1, from the same artifacts as Table 3. The equal-valuation cell is unpointed in its single comet.

(our first $n = 6$ attempt aborted on a detected overflow after six minutes), whereas powers of two reach only t^{32}.

(b) The dependent symmetric equations are essential. Every S2 cell at the five other valuations has a larger f-vector in every entry than its S1 counterpart, the S2 cell at repeated valuations has a different shape (nine entries against six, with far more cones), and no S2 cell yields a certificate at any valuation, including the one where S1 is globally pointed; the single undecided comet of the heuristic screening grows from 71–95 recession generators in the S1 controls to 241–2224 in S2, and the exact decision proves each of them unpointed (Table 3). The equations that are algebraically redundant (each symmetric equation is the sum of two asymmetric ones) are tropically load-bearing. This is the prevariety-level form of the affine phenomenon of Section 4, and it agrees with the historical record: [7] could not run their method on the six AC equations alone, and [10] kept the redundant family deliberately [V].

(c) $n = 4$ falls in seconds, and the hard case is the equal-valuation one. With valuations $(1,3,9,27)$ and $(1,2,4,8)$ the $n = 4$ prevariety is globally pointed (10 comets each; 4 and 2 seconds), a purely polyhedral replication of generic-mass finiteness for the four-body problem, on the fast path; arithmetic valuations $(0,1,2,3)$ certify comet by comet (9 comets). The equal-valuation case $(0,0,0,0)$ leaves a single comet with 49 generators, which is unpointed, matching the observation of [12] that this specialization is equivalent in difficulty to the original computation of [7] [V].

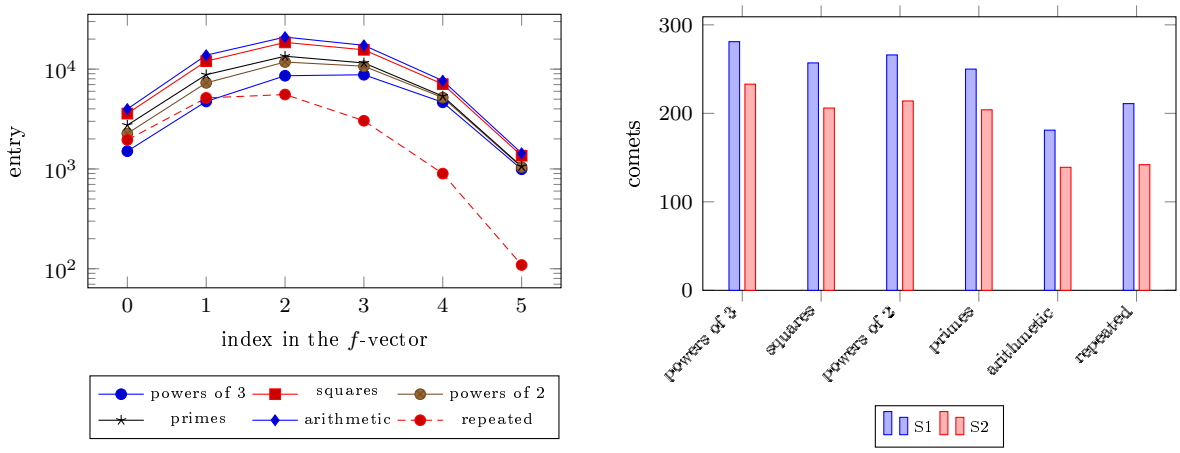


Figure 1: Left: f-vector profiles (entries on a logarithmic scale) of the six S1 cells at $n = 5$, one curve per valuation family. Right: comet counts of systems S1 and S2 by valuation family at $n = 5$. Both panels are generated from the stored artifacts of Tables 3.

6.1 Every comet decided exactly (EXP-007)

On the failing controls the heuristic of the screening returns neither certificate, so those cells read "no certificate" rather than "not pointed". The exact decision of Algorithm 3 closes that gap: a cone is pointed exactly when (2) is infeasible, and both outcomes carry a certificate re-verified by exact substitution outside the decider.

All sixteen cells are decided in 33 seconds total, with zero certificate-verification failures [MV]. Every comet certified pointed by the screening is confirmed (no flips), and each failing control has exactly one comet that is provably *unpointed*: at $n = 5$ the arithmetic valuations $(0, 1, 2, 3, 4)$ (180 pointed, one unpointed) and the repeated valuations, and at $n = 4$ the equal valuations. The data therefore confirm, rather than merely agree with, the failure of arithmetic valuations reported in [12], and localise it to a single component out of 181 [MV].

The logical direction matters: pointedness of every comet is a *sufficient* condition for dimension zero, so an unpointed comet proves only that this certificate route does not close, never that finiteness fails. The prevariety over-approximates the tropical variety [D].

6.2 Adding equations: what helps and what does not (EXP-008)

Since removing the redundant symmetric equations is catastrophic, the converse is worth measuring. At $n = 4$, where the planar stratum *is* the Dziobek stratum and Dziobek's equations are therefore available (they are what Hampton–Moeckel [7] needed when the Albouy–Chenciner equations alone defeated their method), we ran the baseline against three enrichments: the energy–inertia relation e_{IU} (system A2), Dziobek's equations (A3) and both (A4). Table 6 lists the twelve cells; every comet is decided exactly, and the baseline cells reproduce the f-vectors of Table 5 at all three shared valuations.

Two results run against intuition. First, adding equations does not always shrink the f-vector: at equal valuations it shrinks, with the entries falling from $(49, 66, 18)$ to $(37, 48, 12)$ (reductions of 24, 27 and 33 percent), but it *grows* it at powers of two (entries $119, 82, 19$ to $131, 101, 27$) and at arithmetic valuations ($164, 108, 23$ to $183, 135, 34$) [MV]. The prevariety as a set must not be confused with the subdivision that represents it: intersecting more tropical

Valuations	System	comets	pointed	unpointed	gfan (s)	f-vector
$(0,0,0,0)$ equal	A1, baseline	1	0	1	13.2	`1 49 66 18`
$(0,0,0,0)$ equal	A2, $+e_{IU}$	1	0	1	17.9	`1 49 66 18`
$(0,0,0,0)$ equal	A3, + Dziobek	1	0	1	17.6	`1 37 48 12`
$(0,0,0,0)$ equal	A4, + both	1	0	1	17.1	`1 37 48 12`
$(1,2,4,8)$ powers of 2	A1, baseline	10	10	0	–	`63 119 82 19`
$(1,2,4,8)$ powers of 2	A2, $+e_{IU}$	10	10	0	–	`63 119 82 19`
$(1,2,4,8)$ powers of 2	A3, + Dziobek	7	7	0	–	`64 131 101 27`
$(1,2,4,8)$ powers of 2	A4, + both	7	7	0	–	`64 131 101 27`
$(0,1,2,3)$ arithmetic	A1, baseline	9	9	0	66.3	`83 164 108 23`
$(0,1,2,3)$ arithmetic	A2, $+e_{IU}$	9	9	0	92.2	`82 162 107 23`
$(0,1,2,3)$ arithmetic	A3, + Dziobek	6	6	0	71.5	`86 183 135 34`
$(0,1,2,3)$ arithmetic	A4, + both	6	6	0	80.6	`85 181 134 34`

Table 6: The enrichment grid at $n = 4$ (twelve cells; all runs in arbitrary precision). Every comet is decided exactly. The `gfan` seconds are those stored in `table.json` of the experiment; the powers-of-two cells are read from the stored `gfan` outputs and their wall times are not stored (marked "–").

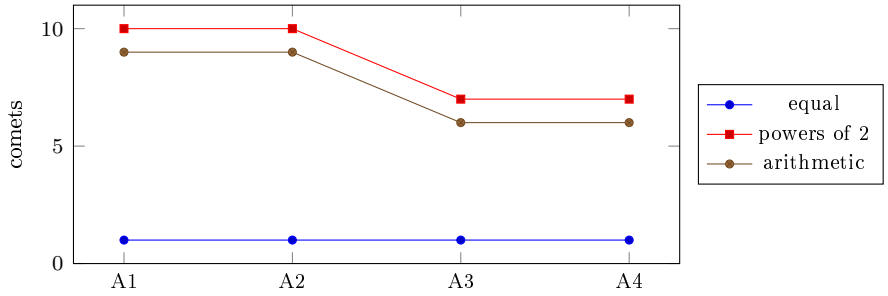


Figure 2: Comet count along the enrichment (A1 baseline, A2 with the energy–inertia relation, A3 with Dziobek's equations, A4 with both) at the three $n = 4$ valuations, generated from Table 6.

hypersurfaces can only shrink the set while refining the fan, so cone counts may rise [D]. The invariant that behaves monotonically in our data is the comet count, which falls from 10 to 7 and from 9 to 6 when Dziobek is adjoined (Figure 2). Second, Dziobek's equations do not rescue the hard equal-valuation case: the prevariety entries fall as above and its single comet stays provably unpointed. The enrichment that rescued the *algebraic* proof therefore does not rescue the *tropical* certificate at the same specialisation, which separates two roles that are easy to conflate. The cells with Dziobek cover the planar noncollinear stratum only (the collinear classes are classical); the positive cells of A3 and A4 are not claimed as full certificates, since the A1 cells already provide those.

By contrast the energy–inertia relation is tropically inert: it reproduces the baseline f-vector exactly at two of the three valuations (equal and powers of two) and shrinks it slightly at the arithmetic one ($82, 162, 107, 23$ against $83, 164, 108, 23$). Dependence in the ideal predicts tropical relevance in neither direction, which is the sharper form of the lesson above, and it lets us drop that equation from expensive runs at no cost [MV].

Experiment	`gfan` arithmetic	Result that rests on it
EXP-003 ($n = 5$, two valuations)	`--bits 64` (fast path)	f-vectors of [12], pointed cone for powers of 3
EXP-004 ($n = 4, 5$, 16 cells)	`--bits 64`	Tables 3 and 5, working families, S2 failure
EXP-008 ($n = 4$, 12 cells)	`--bits 0` (arbitrary precision)	Table 6
EXP-005 ($n = 6$, 0.7)	`--bits 64`; arbitrary precision	overflow abort; internal assertion

Table 7: Arithmetic mode of the prevariety runs.

7 Bit-width provenance and what the certificates rest on

The word *certified* in the screening and reproduction sections applies to the exact decisions on comets, not to the polyhedral computation. Table 7 states which arithmetic mode produced which result.

The overflow-checked rerun of the $n = 5$ computation that [12] describe (about ten times slower) was not repeated, and the full arbitrary-precision rerun of the positive $n = 5$ cells was queued as hardening and no record shows that it ran. The $n = 5$ certificates therefore rest on the fast path alone [D]. Two facts bound the risk: on the fast path an arithmetic overflow is detected and aborts the run, so a silent wrong answer is not the failure mode (the $n = 6$ attempts of Section 10 aborted in exactly this way) [D]; and the arbitrary-precision run of the enrichment grid reproduces the $n = 4$ f-vectors of the fast-path screening exactly at the three shared valuations, $(1, 49, 66, 18)$, $(63, 119, 82, 19)$ and $(83, 164, 108, 23)$ [MV]. That agreement is a partial validation at $n = 4$ only. The arbitrary-precision rerun of the $n = 5$ positive cells is an open item (Section 12).

8 Censuses across two independent engines (EXP-006)

Two of the $n = 3$ censuses do not finish within a 900-second limit in our own engine. We recomputed them with msolve (Gröbner bases, rational univariate representation and certified real-root isolation), keeping our exact machinery as the verification layer: msolve returns rational isolating boxes, and containment of an exact algebraic point in a rational box is decided exactly. Both open cells close in under a second each, and for all four mass vectors the positive census is exactly the classical answer, one equilateral point and one collinear point per ordering, with every one of our independently computed exact points inside a distinct box and no unexplained boxes [MV]. Two implementations that share no code agree on every decided instance.

The same computation shows that the enriched planar $n = 4$ system does not admit an isolated census to compare with the published equal-mass class count: msolve reports the affine system to have dimension one, so there is no isolated census to compare [MV]. Diagnostics exclude the vanishing-distance loci as the cause, which leaves the reading the literature already acts on: the census belongs in the torus, where all mutual distances are invertible, which is precisely the setting of [7] [D]. We record the count as untested rather than quoting the published value as if we had reproduced it.

Following that lesson, we attempted the $n = 4$ equal-mass census where it belongs, in the torus, by two independent exact formulations: the enriched planar system under a Rabinowitsch

saturation of the distance product (seven unknowns) and the z-system of [7] (twelve unknowns, the mutual distances joined by the Dziobek variables). The exact square of Section 3 satisfies the first system, checked by exact residual substitution [MV]. Neither formulation produced any msolve output within a mechanically enforced time limit of 3600 seconds per route (EXP-009), so the published equal-mass counts remain untested by us. We note that the literature contains, to our knowledge, no direct algebraic solve of this census either [U]: [7] bound the count by a mixed volume rather than solving, and the rigorous equal-mass counts of [13] come from interval methods. The measured cliff between $n = 3$ (both censuses close in under a second) and $n = 4$ (no output in an hour, twice) points counting towards bounds rather than solutions, and finiteness towards dimension computations, as follows.

Dimension of the $n = 4$ Dziobek variety (EXP-010). The object is the Dziobek variety of $n = 4$: the vanishing of the cleared Dziobek differences

$$S_{12}S_{34} = S_{13}S_{24} = S_{14}S_{23} \tag{5}$$

together with the planar Cayley–Menger determinant, in the torus of the six mutual distances. Masses do not appear in this formulation, and generic finiteness corresponds to this variety having the dimension of the projectivized mass space, three. The instrument keeps every algebraic call zero-dimensional by construction: recorded random integer linear sections of codimension three (which should meet the variety in finitely many points, whose number is degree data) and codimension four (which should miss it), decided by msolve under caps, with a deterministic Gröbner staircase computation run separately. A validation test discriminates three exact anchors in one second: the exact square of Section 3 lies on the variety (certified by polynomial reduction modulo $8a^3 = 4 + \sqrt{2}$ written as two polynomial relations, no radical heuristics), the unit tetrahedron satisfies the Dziobek equations but is excluded by Cayley–Menger, and the 3-4-5 rectangle satisfies Cayley–Menger but violates the Dziobek equations [MV].

The probes show an asymmetry that is itself the finding. Two independent codimension-four random sections (coefficients drawn uniformly from $[-10^6, 10^6]$ by a seeded generator, the draws archived verbatim) came back *empty* in one second each (msolve's dimension -1 output), which is probabilistic-exact support for the Dziobek variety having dimension at most three, exactly the value the generic finiteness picture predicts and exactly the direction such arguments consume (what must be excluded is excess dimension) [MV]. By contrast, both codimension-three census probes and a deterministic Gröbner staircase computation did not finish within their time limits (900 and 1800 seconds), so no degree data and no dimension lower bound beyond the exact witness were produced, and the census half of the instrument is not pursued at this size. Since only an a-priori degree ceiling is available (Bézout with the stripped degrees $9 \cdot 9 \cdot 6 = 486$), the emptiness result carries a failure estimate of order 10^{-3} per draw; it is evidence, not proof [D]. For the upper-bound direction the natural tools are cheap emptiness probes and the partial-Gröbner device of Dias–Pan [14] (dimension bounds from the union of leading monomials of tractable subideal bases); certified witness sets serve the lower-bound direction. The next case is the spatial five-body Dziobek variety, following the proof shape of [14] with exact instruments.

The scaling test at $n = 5$ (EXP-011). At $n = 5$ the same instruments meet a computational wall. The object is the fifteen cleared Dziobek product differences over all quadruples of five bodies together with the spatial Cayley–Menger equation, in the torus of the ten mutual distances; the cut is an overvariety of the spatial Dziobek central configurations (degenerate strata satisfy the products trivially, as an exact point of the validation set shows: the unit-side, unit-slant bipyramid with $r_{45}^2 = 8/3$, whose membership reduces to rational arithmetic).

The three-anchor validation passed in four seconds; every msolve section probe, including the codimension-five emptiness probes that cost one second at $n = 4$, capped at 300 seconds; and a run of the partial-Gröbner instrument (fifteen subideals, each one product plus Cayley–Menger plus the saturation, 120 seconds apiece under sympy) completed a single subideal, leaving a vacuous union bound [MV]. Nothing refuted the expected dimension four; nothing supported it either. We read the wall as an engine fact rather than a mathematical one: [14] report completing the identical leading-term pattern on an eleven-variable system in minutes under Singular, which motivates a change of Gröbner engine [V].

Engine comparison and the $n = 5$ dimension (EXP-012, EXP-013, EXP-014). Three further computations settle the engine question and reduce the $n = 5$ question to a single open decision. First, the wall is an engine effect: Singular over $\mathbb{Q}$ (release and binary hash in Table 1) completes twelve of the identical fifteen subideal jobs at 0.4–1 seconds each, against one completion near 100 seconds under sympy [MV]. An exact-reproduction control also shows that sympy's Gröbner output uses lexicographic ordering whatever order is requested, so the leading terms harvested in the EXP-011 run were lex leads of grevlex elements; that harvest gave only a vacuous bound, and no result depends on it. msolve's basis mode is modular, so it is used for screening only: an unlucky prime can enlarge a leading ideal and fake a *lower* dimension bound. Second, the union bound gives dimension at most 5, and the fifteen-product system without Cayley–Menger terminates completely in nine seconds, so its staircase dimension is the true Krull dimension: *exactly* 5, matching the rank-one parametrization count [MV]. The cost law is sharp: every computation without Cayley–Menger runs in seconds, every mix of the products with Cayley–Menger caps. Third, the incremental extension of the completed products basis by Cayley–Menger (whose normal form against that basis is nonzero, so the cut is a proper subvariety) also did not terminate within 1800 seconds. The results for the spatial five-body Dziobek cut are therefore: products variety of dimension exactly 5, cut of dimension at most 5, expected value 4 undecided. The remaining decision (does Cayley–Menger vanish identically on a top component?) calls for certified witness sets, and the cost law, measured twice, guides the work on the symmetric strata below.

9 The two-pair symmetric stratum of six bodies (EXP-015 to EXP-023)

Dias–Pan [14] proved generic finiteness for the *cross* symmetric class of the planar six-body problem (four bodies on the reflection axis, one mirror pair) [V]. We study the neighbouring class, with two bodies on the axis and *two* mirror pairs. A targeted search found no published finiteness result for this class; no MathSciNet or zbMATH search was made [U]. This section states what is proved, by what argument, and what is open, under the claim rule of the Introduction. The result is a status report: no finiteness statement is claimed for the stratum.

9.1 The object and its reduction

Bodies 1 and 2 sit on the axis at $(0, a_1)$ and $(0, a_2)$; pair A is $(\pm u, v)$ and pair B is $(\pm p, q)$, with $u, p > 0$. The gauge fixes translation and scale, $a_1 = 1$, $a_2 = -1$. The open stratum $\mathcal{S}$ consists of collision-free configurations with distinct pair heights, $f := v - q \neq 0$. The exact quotient derivation gives nine distance variables (r_{12}; $d_{1A}, d_{1B}, d_{2A}, d_{2B}$; $w_A = 2u$, $w_B = 2p$; $c_s = r_{35}$,

$c_x = r_{36}$) [MV] and the relation

$$c_x^2 - c_s^2 = w_A\, w_B, \tag{6}$$

from $(u+p)^2-(u-p)^2 = 4up$ [MV]. With all six masses free the Laura–Andoyer equations of the two pairs evaluate exactly to

$$L_{34} = -2u\,(m_5-m_6)(q-v)\,\frac{c_x^3-c_s^3}{c_s^3c_x^3}, \qquad L_{56} = +2p\,(m_3-m_4)(q-v)\,\frac{c_x^3-c_s^3}{c_s^3c_x^3}. \tag{7}$$

The identities (7) are machine-verified [MV]. On $\mathcal{S}$ one has $q \neq v$, $u, p \neq 0$ and, by (6), $c_x \neq c_s$, so $L_{34} = L_{56} = 0$ forces $m_5 = m_6$ and $m_3 = m_4$: the masses within each pair are equal (the pair-equality statement) [D]. The factorization (7) is a machine identity; the consequence is its immediate reading. The statement is limited to $q \neq v$: on the sub-stratum $q = v$ both equations vanish identically and force nothing, so that sub-stratum is outside the scope. An attack-based review of the claims records the pair-equality statement as *not tested*: the informative attack, which keeps independent masses on all four pair bodies, lets the geometry vary and solves ten equations in ten unknowns, was not run, and an earlier attack that varied m_A against m_B tested a proposition nobody made (`CC/context/2026-08-25-adversarial-audit.md`, sections 5 and 6). No lemma environment is used for the pair-equality statement for that reason.

By the reflection (34)(56), which preserves all distances and reverses orientation, L_{12}, L_{34}, L_{56} vanish and the other twelve equations pair up with signs, leaving six independent equations $\{L_{13}, L_{15}, L_{23}, L_{25}, L_{35}, L_{36}\}$, each linear in (m_1, m_2, m_A, m_B) [D]; the pairing is confirmed at a witness geometry, where the six partner equations are exact negatives of the block equations [MV]. The block is a 6×4 matrix $J(u, v, p, q)$ whose kernel gives the admissible masses. The gauged shape variety X has dimension 4 (five ungauged), by two engines in one second each [MV]. Its closure X^+ of the image of the irreducible parameter space (a_1, a_2, u, v, p, q) is irreducible of dimension 4 [D]. The matrix has rank 4 at the two exact witnesses $W_1 = (a_1, a_2, u, v, p, q) = (3, -1, 2, 1, 1, -2)$ and $W_2 = (2, -2, 1, 2, 3, -1)$ [MV]. The equal-mass regular hexagon lies in the stratum through the block (all six equations vanish exactly), and its dihedral symmetry lowers the rank to exactly 3 (a nonzero 3×3 minor of value $147/128 - 735\sqrt{3}/512$, all 4×4 minors zero) [MV]. The complete catalogue of the 2×2 minors, computed with the inverse-cube differences $s(a, b) = a^{-3} - b^{-3}$ as abstract symbols, has 84 nonzero entries of the 90 possible, a count reproduced by an independent count over six random shapes [MV].

9.2 The fibre-dimension count

Let $R_j = \{\operatorname{rank} J \leq j\} \cap X^+$ and $I = \{(x, m) : x \in X^+,\ m \in \ker J(x),\ m \neq 0\}$ with projection $\pi : I \to \mathbb{R}^4$ to mass space. Over the locus of rank exactly r the fibre of $I \to X^+$ is the kernel, of dimension $4 - r$, so

$$\dim I = \max_r \big[\dim\{\operatorname{rank} J = r\} + (4 - r)\big]. \tag{8}$$

Proposition 9.1. *If* $\dim R_0 \leq 0$, $\dim R_1 \leq 1$, $\dim R_2 \leq 2$ *and* $R_3 \neq X^+$, *then* $\dim I \leq 4$.

Proof. Term by term in (8): for $r = 4$ the term is at most $4 + 0$; for $r = 3$, irreducibility of X^+ and $R_3 \neq X^+$ give $\dim R_3 \leq 3$, so the term is at most $3 + 1$; for $r = 2$ it is at most $2 + 2$, for $r = 1$ at most $1 + 3$ and for $r = 0$ at most $0 + 4$. □

From $\dim I \leq 4 = \dim \mathbb{R}^4$ the projection π has finite generic fibres, or is not dominant, in which case generic masses admit no configuration of the stratum at all; this is the fibre-dimension argument of the Dias–Pan pattern [14] (the record's section 2 of `CC/context/2026-08-`

k	Statement	Status	Written proof in the records
0	R_0 misses $\mathcal{S}$	proved [D] [MV]	Lemma 9.3 below; `CC/experiments/EXP-022-collar-coverings/r0-lemma.py` (docstring)
1	$\dim R_0 \le 0$	proved [D] [MV]	piece 5, `CC/context/2026-08-02-rank-floor-lemma-dossier.md` lines 199–236
2	$\dim R_1 \le 1$	proved [D] [MV]	pieces 1 to 4, same file, lines 8–155 (Lemma 9.2 gives piece 1)
3	$\dim R_2 \le 2$	reduced to four bilinear conditions [D]	piece 7b, same file, lines 313–341; certified covering below
4	$R_3 \neq X^+$	proved [D] [MV]	Proposition 9.4 below; same file, lines 270–289

Table 8: Status of the cases of the ladder. A case is called proved only with a written proof in this paper or at the cited lines.

`20-stratum-theorem-assembly.md`, lines 26–62). The four hypotheses are the *ladder*. Following the case numbering of the records, case k is the statement $\dim R_{k-1} \le k-1$ (for $k = 1, 2, 3$) and $R_3 \neq X^+$ (for $k = 4$). The centred pentagon below shows that $\dim R_2 = 2$ is attained, so the bound for $k = 3$ is sharp.

9.3 The proved cases and their pointers

Lemma 9.2 (piece 1). *On $\mathcal{S}$ the 2×2 minor of J on rows $\{L_{35}, L_{36}\}$ and columns $\{m_1, m_2\}$ equals*

$$M = s(d_{1A}, d_{1B})\, s(d_{2A}, d_{2B}) \big(-2\, u\, p\, (v-q)(a_1 - a_2)\big). \tag{9}$$

Hence $\operatorname{rank} J \ge 2$ *on $\mathcal{S}$ outside $E = \{d_{1A} = d_{1B}\} \cup \{d_{2A} = d_{2B}\}$.*

Proof. The four entries are single terms of the form $s(d_{iA}, d_{iB})\,\Delta$, with the same s-factor in each column, so the s-factors come out of the determinant. The polynomial bracket $\Delta_{351}\Delta_{362} - \Delta_{361}\Delta_{352}$ equals $-2up(v-q)(a_1-a_2)$ identically (an expanded residual that is the zero polynomial; by hand, $\alpha\gamma - \beta\delta = -(v-q)(a_1-a_2)$ in the height variables). On $\mathcal{S}$ the bracket is nonzero because $u, p > 0$, $v \neq q$ and $a_1 \neq a_2$. Where it is nonzero the minor is nonzero unless an s-factor vanishes, which is the set E. □

Lemma 9.3 ($k = 0$). *$R_0 \cap \mathcal{S} = \emptyset$.*

Proof. The entries $J[L_{23}][m_1] = s(r_{12}, d_{1A})\, u\, e_{12}$ and $J[L_{13}][m_2] = -s(r_{12}, d_{2A})\, u\, e_{12}$ (and the analogous entries for pair B) are single s-terms with nonvanishing monomial brackets on $\mathcal{S}$. If $J = 0$ then $d_{1A} = d_{2A} = r_{12} = 2$ and $d_{1B} = d_{2B} = 2$. From $d_{1A}^2 - d_{2A}^2 = -4v$ one gets $v = 0$, and likewise $q = 0$, so $f = v - q = 0$, contradicting $q \neq v$ on $\mathcal{S}$. □

Proposition 9.4 ($k = 4$). *$R_3 \neq X^+$, hence $\dim R_3 \le 3$.*

Proof. At W_1 the matrix has rank 4, by exact radical arithmetic, so some 4×4 minor g_4 satisfies $g_4(W_1) \neq 0$ exactly. The point W_1 is a parametrized geometry, hence a point of X^+. On R_3 every 4×4 minor vanishes, so $R_3 \subset \{g_4 = 0\} \cap X^+$. Since X^+ is irreducible of dimension 4 and g_4 does not vanish identically on it, Krull's principal ideal theorem gives $\dim(\{g_4 = 0\} \cap X^+) = 3$. □

The cases $k = 1$ and $k = 2$ combine closed-form minors with exact witnesses. Four minors of the catalogue factor into monomials times products of the symbols $x_i = s(d_{iA}, r_{12})$ and

$y_i = s(d_{iB}, r_{12})$; the Boolean analysis of the four products shows that $\operatorname{rank} J < 2$ forces $x_1x_2 = y_1y_2 = y_1x_2 = x_1y_2 = 0$, hence one of two branches ($d_{1A} = d_{1B} = r_{12}$ or $d_{2A} = d_{2B} = r_{12}$), each of codimension two. On each branch the mixed-column minors add a third condition drawn from five explicit degenerations, each proper because the 3-4-5 witness $(a_1, a_2, u, v, p, q) = (5, 0, 3, 4, 4, -3)$ (and its mirror) lies on the branch and violates all five; whence $\dim R_1 \le 1$ (piece 4). For $k = 1$ the two height sub-branches are empty by one-line substitutions, and the remaining cuts are proper by witnesses over $\mathbb{Q}(\sqrt{3})$, giving $\dim R_0 \le 0$ (piece 5). Every identity is verified exactly in the records, and the 84-minor catalogue is the input of the Boolean analysis. We do not reproduce those two proofs here; Table 8 points to their lines, and the status of each is stated by that pointer.

9.4 The case $k = 3$: reduced, not closed

On the region where the corner minor $\{L_{13}, L_{23}\} \times \{m_1, m_2\}$ is nonzero (off $\{d_{1A} = r_{12}\} \cup \{d_{2A} = r_{12}\}$), the eight borders of that corner factor, with all four identities verified exactly, and R_2 lies in the common zero set of four equations, each with at most five s-monomial terms:

$$b_{13}J[L_{15}][c] = b_{15}J[L_{13}][c], \qquad a_{23}J[L_{25}][c] = a_{25}J[L_{23}][c], \qquad c \in \{m_A, m_B\}, \tag{10}$$

where $a_{23} = s(r_{12}, d_{1A})\, u\, e_{12}$, $b_{13} = -s(r_{12}, d_{2A})\, u\, e_{12}$, $b_{15} = -s(r_{12}, d_{2B})\, p\, e_{12}$ and $a_{25} = s(r_{12}, d_{1B})\, p\, e_{12}$ are the clean entries [MV]. All four conditions are exactly nonzero at W_1, W_2 and the hexagon, so by irreducibility and Krull each cuts X^+ in dimension 3, and $\dim R_2 \le 3$ follows on the good region; the case needs $\dim R_2 \le 2$, that is a second cut that is not identically zero on any 3-dimensional component of the first, and that statement has no written proof in the records [D]. The routes tried to close it are the computational ones of Table 10: the single-minor Gröbner cuts (EXP-019), the sign-chamber relaxation (EXP-020) and the certified covering (EXP-021, EXP-022). The covering does not prove the case by itself, because its statement is not assembled (below).

9.5 The certified interval atlas

A box of the coordinates (u, v, p, q) is discharged by one of three certificates, all in exact rational interval arithmetic with outward rounding: (a) *rank* ≥ 3, some 3×3 minor of J excludes zero over the box; (b) *trap*, a 2×2 minor excludes zero over the box (so R_1 misses it) and two 3×3 minors have a 2×2 gradient subdeterminant excluding zero (so R_2 meets it inside a smooth codimension-2 manifold, dimension 2); (c) bisection to a declared depth cap, with residual failures investigated rather than assumed away. Where a limit is singular (collisions, pair collapses, infinity) the matrix is first put in an analytic chart, with every singular factor cancelled algebraically, and each chart is gated by a five-point crosscheck against the original matrix. Symmetry gives a Klein four-group of exact matrix identities (the swap $(u, v, p, q) \mapsto (p, q, u, v)$ and the mirror $(v, q) \mapsto (-v, -q)$) under which every certificate is invariant [MV].

Residual failures of the charts have three origins. A box that contains an excluded face (a collision locus or a blow-up face) cannot be certified, because the matrix genuinely drops rank there. The face principle resolves it in two halves: every dyadic shell at positive distance from the face certifies (measured over six halvings), and closed forms make the rank floor uniform over the shells. Piece 10 treats the corner face (a pair on an axis body), piece 11 the pair-collapse face (both pairs onto the axis, with explicit leading coefficient $4ab\,[\varphi(1-v)\varphi(-1-q) - \varphi(-1-v)\varphi(1-q)]$, $\varphi(x) = x(1/8 - 1/|x|^3)$) and piece 12 the two merge faces [MV]. Limited resolution

Residue	Boxes	Disposition
15 charts	0 failures	complete
fa2b	200	discharged at depth 76: 1544 boxes processed, zero failures
cb1, cb1f	5 and 2	all touch a collision face, closed by piece 10
band	44 (and 48)	the cross point: a rank-2 degenerate configuration; all 92 boxes are trapped, so $\dim R_2 \leq 2$ holds there
m2-L, m2-R	5824 and 5683	the quadruple corner: a chart region still open; 6208 boxes of each chart are covered by the collapse chart, the rest by no chart
fa2, m1vert-N	retired	replaced by fa2b and by per-entry guards

Table 9: Residue accounting of the atlas of the stratum `CC/context/2026-08-20-stratum-theorem-assembly.md`, sections 6b, 6c, 8 and 9.

is the second origin: for every chart with failures, the failures share exactly one box width, the width reached at the shared depth cap, and a deeper cap discharges them [MV]. The third origin is mathematics, once: the cross point below. The face table of the atlas is complete: every chart's matrix evaluated on its own face is either full rank or closed by pieces 10 to 12 [MV]. Table 9 gives the residue accounting.

The atlas therefore leaves one mathematical residue, the cross point (next subsection), and one open chart region, the quadruple corner of the charts m2-L and m2-R. The corner is a defect of the chart and not of the mathematics: the matrix has rank 3 throughout the open stratum there, the m_1 and m_2 columns are $O(s)$ while the m_A and m_B columns are $O(s^{-2})$ (a ratio of exactly s^3), and about half of the residual boxes straddle the physical boundary $u_h, p_h \geq 0$, which is not axis aligned. The repair is specified (reparametrize the corner by (u_h, p_h) and divide the two axis-mass columns by their common factor), and the records do not show it carried out. A first check of the corner compared the box coordinates of two charts that do not share a parametrization and reported a hole that was not there; the corrected comparison, through the exact map $c_s = R_c\, s_t$, covers 6208 of the boxes of each chart and leaves the remainder to a delegation loop between the charts [MV]. So the quadruple corner remains open and the covering is complete except for it.

9.6 The cross point and the finiteness of its family

The residue of the band chart sits at $v = q = 0$, with both mirror pairs at height zero (the doubly symmetric cross). On that face σ_4 vanishes identically, so the rank is 3 and each point has a unique mass ray; the cross face carries a two-parameter family of central configurations. At $u^* = 0.6309181371\ldots$, $p^* = 1.4509074659\ldots$ the rank is exactly 2: σ_3/σ_1 falls in proportion to the search bracket over thirteen orders of magnitude, and a 3×4 system in (m_1, m_A, m_B, λ) built from scratch, sharing no code, confirms $\sigma_3/\sigma_1 = 8.6 \times 10^{-34}$ against controls at 8.6×10^{-3}, 6.6×10^{-3} and 5.8×10^{-2} [MV]. The two-dimensional kernel meets the positive orthant in an arc of width 1.2 rad, so one configuration is central for a one-parameter family of positive masses: a degenerate central configuration, with smallest mutual distance 0.82. A census over the face (5607 grid points, seven basins) finds exactly one such point [MV].

Symmetry reduces the equations to three scalar conditions, each linear in (m_1, m_A, m_B, λ). Eliminating the algebraic half-powers $C = (u^2+1)^{-3/2}$ and $D = (p^2+1)^{-3/2}$ by resultants leaves

two integer polynomials g_1, g_2 in (u, p) of 2027 and 936 terms, both vanishing at the point to relative 3.6×10^{-60} and 2.6×10^{-65} [MV]. The polynomial g_1 is the square of an irreducible polynomial of total degree 40, g_2 is irreducible of total degree 56, and $\gcd(g_1, g_2)$ has total degree 0 [MV]. The common zero set is therefore zero-dimensional: the degenerate configurations of the doubly symmetric cross family are finite, at most $40\times56 = 2240$ over $\mathbb{C}$ with multiplicity (the resultants may carry extraneous factors, which places the true set inside their common zero set, the direction finiteness needs) [MV] [D]. The written argument is in `CC/context/2026-08-02-rank-floor-lemma-dossier.md`, lines 1227–1272, and `CC/history/log.md`, lines 822–880 (Round 54), with the factorizations and the greatest common divisor computed exactly; the statement is not given a theorem environment here because the irreducibility and coprimality computations are not reproduced in the text. A numerical identification by integer relation search found nothing real and was rejected. The scope is the doubly symmetric sub-family (line masses m_B, m_A, m_A, m_B, off-line masses equal), and the statement concerns the *degenerate* configurations, a different question from the one Dias and Pan answer. The family is not new: it lies in their cross class [V], but their theorem excludes a non-generic mass set, and the configuration is an explicit witness in it (their Proposition 5.2 example is a different configuration). Whether this specific configuration appears in the wider literature on symmetric central configurations is not checked [U].

The centred regular pentagon (bodies $1, 3, 4, 5, 6$ on a regular pentagon of circumradius 2 centred at body 2) is an exact rank-2 point with kernel spanned by the free central mass and the equal ring masses, found from interval failures whose count stayed constant under bisection; with exact gradients of two 3×3 minors of rank 2 at that point (piece 8, by the implicit function theorem) R_2 is locally a codimension-2 submanifold near it [MV].

9.7 The dropped statement and the open questions

A draft statement for the stratum named the centred pentagon as the sharp degenerate point of the open stratum, and asserted finiteness of the configurations for all masses outside a proper closed set. It was dropped on 23 August 2026 because it is false as written: the cross point is a second exact rank-2 point with an admissible mass ray, found on the face $v = q = 0$, which the declaration of the open stratum excludes, so the uniqueness wording fails. The statement is not repeated here. Three questions have to be settled before any statement is rebuilt: (1) whether the cross point is the only degenerate point on the closure of the stratum, or only on the cross face (the census covered the cross face alone, in a bounded window); (2) whether the open stratum, with pairs at strictly distinct heights, contains a degenerate point besides the centred pentagon; (3) whether the cross point is known in the literature [U]. These are open. The atlas, the cases $k = 0, 1, 2, 4$ and the reduction of $k = 3$ are the established parts; the finiteness statement for the stratum is not established.

9.8 The stratum with three mirror pairs (EXP-023)

The mirror symmetry kills L_{12}, L_{34}, L_{56} and pairs off the rest, leaving six independent equations over three masses (a 6×3 matrix); its generic rank is 3 (20 of 20 random shapes), so a generic shape admits no masses and the central configurations of the stratum lie on the rank-at-most-2 locus, a codimension-2 subvariety of the four-dimensional shape space. The regular hexagon has rank exactly 2 with the equal-mass kernel, which validates the instrument. Unlike the two-pair stratum, the face matrices are full rank (pair collapse at 27 of 27 sample points, pair merge at 40 of 40, the all-narrow region flat at $\sigma_3 = 2.0$), so no face lemma and no chart at infinity is needed.

A compact gauge (translation $v_1 = 0$ and scale $\max(u_1, u_2, u_3, |v_2|, |v_3|) = 1$) replaced two charts that exchanged their residues across arbitrary boundaries. Six collinear bodies span no triangle, so the Laura–Andoyer matrix is identically zero on the interior collinear locus, which only this stratum of the family has and which has dimension 2; the equations are vacuous there, and the classical count of Moulton [17] settles the collinear shapes, one positive mass ray each [D]. A lemma piece (piece 13) gives the merge-face rank floor, which is not bounded below; the merge residue then lies on the surface where one entry vanishes. The stratum still has its collapse and merge faces open (14,485 collapse failures, all face-touching) and its blow-up sub-chart unbuilt [MV].

10 The $n = 6$ attempt (EXP-005)

The screening of Section 6 turns the $n = 6$ question into a design problem with two dials. What we have measured is a *tooling* barrier, characterized on four attempts with `gfan` 0.7. On the machine-integer fast path the computation aborts within minutes on a *detected* overflow (the benign failure mode: `gfan` raises rather than returning a wrong answer), and it does so for powers of three and for powers of two alike, although the latter caps the valuation exponents at 32 instead of 243, so the failure is not a matter of valuation magnitude. On the arbitrary-precision path, the powers-of-two run died after fifteen wall-hours (about six cpu-days) on an internal assertion (`gfanlib_hypersurfaceintersection.cpp:505`) in the hypersurface-intersection code [MV]. `gfan` 0.7 therefore fails at $n = 6$ in both arithmetic modes, in two distinct ways, while the same binary reproduces the published $n = 5$ certificates exactly; [12] report their own $n = 6$ computation as completing (and inconclusive on pointedness) [V], so their run differed in version, flags, or valuation. We rebuilt the pipeline on the 0.8beta release, validated it against our $n = 5$ control (exact f-vector agreement), and confirmed that it clears 0.7's first failure point [MV].

The 0.8beta runs have not completed. The powers-of-two run on the fast path (`--bits 64`) has been paused since 20 August 2026 to free processors for the coverings of Section 9; it is protected by a 1.2 GB checkpoint written with `--saveas`, so the pause costs only the work since the last checkpoint. The powers-of-three run with arbitrary-precision integers (`--bits 0`) is parked after four launches, none of which wrote a checkpoint (`CC/experiments/EXP-005-n6-prevariety-pow3/PAUSE-NOTE.md`) [MV]. The measured quantities (fifteen wall-hours, the assertion location) are kept only in the history log of the problem, as the experiment folder has no result file. The case $n = 6$ therefore remains open here: a completed run would either extend generic-mass finiteness to $n = 6$ or localize the obstruction to specific comets, data that has not been published for any failed attempt. Resuming it is the only open case of the programme, and needs a fresh hypothesis and a budget before any run.

11 The five-body case for all positive masses: claims and a formal audit (EXP-024)

In October 2026 three groups claimed that the planar five-body problem has finitely many central configurations up to similarity for every positive mass vector: Tomar [1], with a Lean 4 formalization and a supplement of certificates; Chen and Tong [2]; and Liu and Liu [3] [V]. All three complete the Albouy–Kaloshin classification of singular sequences [8] on the exceptional mass set that [8] leaves open. The question of the previous sections for $n = 5$ is therefore settled, if these claims hold, and the priority is theirs. None of the three is refereed.

We audited the formalization of [1] without rebuilding it. Its main theorem, $\forall m,\ (\forall i,\ 0 < m_i) \to \mathrm{FinitelyManyCC}\ m$, is built from a planar configuration $q : \{1, \dots, 5\} \to \mathbb{C}$ with distinct points, the equation $\lambda m_i (q_i - c) = \sum_{j \neq i} m_i m_j (q_i - q_j)/|q_i - q_j|^3$ with $\lambda \in \mathbb{R}$ and $c \in \mathbb{C}$, and similarity by every map $z \mapsto az + b$ or $z \mapsto a\bar{z} + b$, $a \neq 0$; finitely many central configurations means finitely many similarity classes. This is the classical statement: the sign of the force is absorbed by λ, the value $\lambda = 0$ admits no configuration for positive masses, and counting up to reflections changes counts, not finiteness [D]. The statement was read in the stored sources `System.lean` and `Basic.lean` [V].

A scan of the stored sources and the generated modules (each archive checked against its published SHA-256) searches for nine textual patterns: `sorry`, `admit`, an axiom declaration, `native_decide`, `implemented_by`, `extern`, `ofReduceBool`, `unsafe` and `skipKernelTC`. The 2,751 stored source files were scanned with comments stripped, and the 85,756 generated modules by a raw search including comments. None of the nine patterns occurs in the formalization; the three hits (all `unsafe`) are in the exporter used to feed independent kernels [MV]. The scan is a text scan, with a limit: the axiom pattern matches only at the start of a line after the prefixes `private`, `protected` and `noncomputable`, so an `axiom` preceded by an attribute, or a declaration introduced by a metaprogram, would not match; the output of the `#print axioms` command on the main theorem was not obtained. The scan therefore shows that the theorem as formalized is the right statement and that none of the stored or generated sources contains any of these nine patterns. It does not replace the kernel check, which the author reports for a full build of about 863 CPU-hours and for two independent kernels, and which we did not reproduce. The open case for this programme is $n = 6$.

12 Limitations and open items

1. The prevariety certificates at $n = 4, 5$ of Sections 5 and 6 rest on the machine-integer fast path. The overflow-checked rerun is an open item, not a result; it is not claimed here. Arbitrary precision was used only in the enrichment grid at $n = 4$.

2. The `gfan` 0.7 tarball and binary hashes are not among the committed artifacts, so that part of the provenance (Table 1) is unverified; the hashes of every input and output of the screening grid are.

3. Pointedness of every comet is a sufficient condition for dimension zero of the tropical variety; an unpointed comet proves only that the certificate route does not close. The comet reading depends on `gfan`'s semantics under `--usevaluation`, validated by the count of 257 components.

4. The section probes of Section 8 are probabilistic and see only the maximum dimension. The completeness of `msolve`'s box list is trusted and checked only against the classical counts at $n = 3$.

5. The two-pair stratum of the six-body problem has no finiteness statement. The case $k = 3$ is not proved, the quadruple-corner chart region is open, the pair-equality statement is untested by its informative attack, and three questions on the degenerate points are open.

6. The five-body claims are unrefereed and were not rebuilt; the audit of the Lean statement is a text scan with the limits stated.

7. The enrichment cells at powers of two are read from stored outputs; their wall times were not stored.

13 Scope and data availability

The study is replication-first. The finiteness theorems replicated are due to Hampton–Moeckel, Albouy–Kaloshin, and Jensen–Leykin; the equal-mass censuses to Moczurad–Zgliczyński. Our measurements (the dimension-blindness anchors with their exact minimal polynomials and J-invariants; the enrichment law at affine and prevariety level; the screening tables) are machine facts about known objects, stated with their provenance. The censuses for two integer-separated mass vectors, which our sympy engine does not finish, are closed by an independent engine (`msolve`) with the sympy layer retained as exact verification, while the $n = 4$ torus censuses and the Dziobek census probes remain capped as reported above. Statements that rest on a literature source not read in its primary form carry the mark [U].

The tables and figure data of Section 6 are generated from the stored artifacts by `gen_tables.py` next to the source, which stops when a recomputed cell differs from a stored value. Code, artifacts and experiment records are available at `https://github.com/fsantibanezleal/CAOS_RESEARCH` under `problems/dynamical-systems/central-configurations/`. The identifiers EXP-001 to EXP-024 in the section titles and the text denote the experiment records in its `experiments/` folder; each record holds the code, the artifacts and the result. Tool versions and binary hashes (`gfan`, `msolve`, Singular) are in Table 1 and in the records that use them.

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Experiment	Instrument	Result
EXP-015	Gröbner staircase and Singular `dim()`	shape variety of dimension 5 ungauged, 4 gauged, in one second, two engines agree [MV]
EXP-016	exact radical rank at two witnesses	rank 4 at W_1 and W_2; component decomposition capped [MV]
EXP-017, 017b, 017c	Gröbner bounds for all four loci, s-variable model, mod-p triage	every locus bound capped in all three formulations; all ten mod-p cells capped at two primes, so the Gröbner route to the loci bounds is closed at the budgets tried [MV]
EXP-018	hexagon through the block	hexagon is a stratum configuration with rank exactly 3 [MV]
EXP-019	Singular std on the shape ideal plus one minor (g_4, g_3)	both single-minor cuts capped at 1800 s despite a 32-term increment over a one-second base; all four mod-p cells capped at 300 s, so the Gröbner route to the loci bounds is closed at every granularity tried [MV]
EXP-020	GF(2) inclusion–exclusion over 24 orientation chambers	125,097,984 residual sign assignments remain: the free-atom sign relaxation does not close $k = 3$, so the geometric couplings are essential [MV]
EXP-021	verified covering of the core	854,317 boxes in 5,096 s, zero residual failures; four pentagon balls (radius 2^{-8}) carry both certificates; $\dim(R_2 \cap \text{core}) \leq 2$, $R_1 \cap \text{core} = \emptyset$, $R_0 = \emptyset$ [MV]
EXP-022	collar, tube, collapse, corner and outer coverings; face lemmas 10 to 12	fifteen charts at zero failures; the residues of Table 9; the cross point and its exact algebraic system [MV]
EXP-023	the stratum with three mirror pairs and no axis body	6×3 matrix of generic rank 3, so central configurations lie on the rank-2 locus; hexagon is rank 2 with the equal-mass kernel; every face is full rank and a compact five-chart atlas passes a seam gate (48,358 samples, zero gaps) [MV]; collapse and merge faces still carry face-touching failures, and the blow-up sub-chart is unbuilt

Table 10: Experiments on the symmetric strata of the six-body problem, from the experiment records.